A Central Limit Theorem for the High-Temperature SYK Model

Abstract

Fix an even interaction order $q \geq 4$, and let $Z_N(\beta)$ denote the partition function of the Gaussian Sachdev–Ye–Kitaev model with N Majorana fermions. For every sufficiently small fixed inverse temperature $\beta \neq 0$, we prove that

$$\frac{Z_N(\beta) - \mathbb{E}Z_N(\beta)}{\sqrt{\operatorname{Var} Z_N(\beta)}} \Longrightarrow \mathcal{N}(0,1).$$

We also identify the limiting variance of the relative partition function and obtain the corresponding central limit theorem for the free energy. The leading fluctuation is the radial component of the second Gaussian chaos. The higher chaoses are controlled by combining a sign-gauge parity rule at logarithmic chaos order with analytic suppression of the remaining tail.

Keywords: Sachdev–Ye–Kitaev model, free energy, central limit theorem, Gaussian chaos.

Mathematics Subject Classification (2020): 60F05, 60B20, 82B44.

1 Introduction and main results

The Sachdev–Ye–Kitaev (SYK) model is a mean-field model of randomly interacting fermions originating in the work of Sachdev and Ye [1] and its Majorana formulation by Kitaev. Its combination of solvability, strong interactions, and chaotic behavior has made it a central model in quantum statistical mechanics and holography; see, for example, [2]. At positive temperature, the partition function contains both the thermodynamic information and the sample-to-sample fluctuations of the disorder.

Rigorous results have recently identified the high-temperature annealed and quenched free-energy limits [5] and established complex zero-freeness and second-moment estimates [4]. The purpose of this paper is to determine the leading distributional fluctuations in the same high-temperature regime. We show that the partition function and its logarithm have Gaussian fluctuations at scale $N^{1-q/2}$ relative to the annealed partition function and identify their limiting variance.

1.1 The SYK model

Fix an even integer $q \geq 4$, and let N be even. Write

$$\mathcal{I}_N = \{I \subset [N] : |I| = q\}, \qquad M_N = \binom{N}{q}, \qquad \sigma_N^2 = \frac{(q-1)!}{N^{q-1}},$$

and let $(g_I)_{I \in \mathcal{I}_N}$ be independent standard Gaussian random variables. Let $\psi_1, \dots, \psi_N$ be Majorana matrices satisfying $\{\psi_i, \psi_j\} = 2\delta_{ij}\mathbb{I}$. For $I = \{i_1 < \cdots < i_q\}$ set

$$\psi_I = \mathrm{i}^{q/2} \psi_{i_1} \cdots \psi_{i_q}.$$

Then ψ_I is Hermitian, $\psi_I^2 = \mathbb{I}$, and $\|\psi_I\| = 1$. Define

$$H_N(g) = \sigma_N \sum_{I \in \mathcal{I}_N} g_I \psi_I, \qquad Z_N(z; g) = \operatorname{tr} e^{-zH_N(g)}, \qquad F_N(z) = \mathbb{E}Z_N(z; g).$$

Here $\mathrm{tr} = 2^{-N/2}\,\mathrm{Tr}$ denotes the normalized trace; in particular, $Z_N(0) = F_N(0) = 1$. This normalization does not affect the relative fluctuations studied below. Throughout, all limits are taken as $N \to \infty$ through even integers.

1.2 Main results

Theorem 1.1 (Partition-function CLT)**.** *Let R_q and β_{FE} be the constants in Propositions 2.1 and 2.4, respectively, and let ℓ_q be the limiting annealed free energy in Proposition 2.4. If $\beta \in \mathbb{R}$ satisfies*

$$0 < |\beta| < \min\{\beta_{\mathrm{FE}}, R_q\}, \qquad \beta \ell_q'(\beta) \neq 0.$$

Then

$$N^{q/2-1}\left(\frac{Z_N(\beta)}{\mathbb{E}Z_N(\beta)} - 1\right) \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta)\,\xi, \qquad \xi \sim \mathcal{N}(0,1). \tag{1}$$

Moreover,

$$N^{q-2}\frac{\mathrm{Var}\,Z_N(\beta)}{(\mathbb{E}Z_N(\beta))^2} \longrightarrow \frac{q!}{2}\,\beta^2\ell_q'(\beta)^2, \tag{2}$$

and hence

$$\frac{Z_N(\beta) - \mathbb{E}Z_N(\beta)}{\sqrt{\mathrm{Var}\,Z_N(\beta)}} \Longrightarrow \mathcal{N}(0,1). \tag{3}$$

Corollary 1.2 (Log-partition-function CLT)**.** *Under the assumptions of Theorem 1.1,*

$$N^{q/2-1}\left(\log Z_N(\beta) - \mathbb{E}\log Z_N(\beta)\right) \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta)\,\xi, \qquad \xi \sim \mathcal{N}(0,1).$$

Equivalently, if $\mathcal{G}_N(\beta) = -\beta^{-1}\log Z_N(\beta)$ is the random free energy, then

$$N^{q/2-1}\left(\mathcal{G}_N(\beta) - \mathbb{E}\mathcal{G}_N(\beta)\right) \Longrightarrow -\sqrt{\frac{q!}{2}}\,\ell_q'(\beta)\,\xi.$$

1.3 Relation to earlier work

Feng, Tian, and Wei [3] proved a central limit theorem for linear eigenvalue statistics of the Gaussian SYK Hamiltonian. Their theorem treats a fixed test function f under regularity assumptions on f and f'. In the normalization used in that paper, however, the partition function corresponds to

$$f_{N,\beta}(x) = e^{-\beta\tau_N x}, \qquad \tau_N \asymp \sqrt{N},$$

so both the test function and its derivative vary with N. The fixed-test-function theorem therefore does not apply directly to $Z_N(\beta)$.

Our argument retains the radial-fluctuation mechanism identified in [3], but establishes the uniform remainder estimate needed for this N-dependent sequence. A complex second-moment bound alone is of order $N^{1-q/2}$, whereas the central-limit scale requires an error $o(N^{2-q})$. We bridge this gap by splitting the Gaussian-chaos expansion at order $A\log N$. A sign-gauge counting argument controls all lower orders uniformly, while analyticity produces a geometric factor for the higher-order tail. The proof uses only annealed free-energy convergence and does not assume any fluctuation theorem for either Z_N or $\log Z_N$.

1.4 Outline of the proof

We first relate our normalization to that of Feng, Tian, and Wei. Define

$$\widehat{H}_N(g) = M_N^{-1/2}\sum_{I\in\mathcal{I}_N} g_I\psi_I, \qquad \tau_N = \sigma_N\sqrt{M_N}. \tag{4}$$

Then $H_N = \tau_N \widehat{H}_N$ and

$$\tau_N^2 = \sigma_N^2 M_N = \frac{N}{q}\big(1 + O_q(N^{-1})\big). \tag{5}$$

Consequently, for $\widehat{L}_N(f) = \operatorname{tr} f(\widehat{H}_N)$, then

$$Z_N(\beta) = \widehat{L}_N(f_{N,\beta}), \qquad f_{N,\beta}(x) = e^{-\beta\tau_N x}. \tag{6}$$

Set $a = q/2 - 1$ and normalize the partition function by $Y_N = Z_N/F_N$. Clifford sign-gauge symmetry eliminates the first Gaussian chaos and identifies the second chaos exactly:

$$Y_N(\beta) - 1 = A_N(\beta) Q_N + \mathcal{R}_N(\beta),$$

where

$$A_N(\beta) = \frac{\beta}{2}\partial_\beta \log F_N(\beta), \qquad Q_N = \frac{1}{M_N}\sum_I (g_I^2 - 1).$$

Here $\mathcal{R}_N$ is the sum of the Gaussian chaoses of orders at least three. Thus the leading random term is a normalized sum of M_N independent centered variables. The main task is to prove

$$\|\mathcal{R}_N(\beta)\|_2^2 = o(N^{-2a}).$$

We split the chaos expansion at $m_N = \lceil A \log N \rceil$. Up to order m_N, a Duhamel derivative bound and the sign-gauge parity rule reduce the estimate to counting even tuples of q-sets. Such a tuple of length r uses at most $qr/2$ vertices, which makes the sum of chaoses of orders $3, \dots, m_N$ of order $o(N^{-2a})$. Above m_N, the r-th chaos vanishes to order at least r at $\beta = 0$. An L^2-valued analytic estimate on a zero-free disk, together with the complex second-moment input, gives a geometric factor $(|\beta|/R)^{2m_N}$. Choosing A sufficiently large yields the same $o(N^{-2a})$ bound for the remaining chaoses.

The annealed free-energy input and convexity imply

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2}\ell_q'(\beta).$$

Since $M_N \sim N^q/q!$, the classical central limit theorem for Q_N, the remainder estimate, and Slutsky's theorem give the claimed limit for $N^a(Y_N - 1)$. Orthogonality of distinct Gaussian chaoses gives the matching variance asymptotic and hence the standardized central limit theorem.

2 Analytic and free-energy inputs

2.1 A complex second-moment estimate

We use the following high-temperature estimates, which follow from Theorems 3 and 4 of [4] after adjusting the inverse temperature to our Majorana normalization. All constants in the statement depend only on q.

Proposition 2.1 (Complex second-moment input)**.** *There are constants $R_q > 0$, $C_q < \infty$, and $N_q < \infty$ such that, for every even $N \geq N_q$,*

$$F_N(z) \neq 0 \qquad (|z| \leq R_q), \tag{7}$$

and

$$\sup_{|z| \leq R_q} \mathbb{E}\left|\frac{Z_N(z;g)}{F_N(z)} - 1\right|^2 \leq C_q N^{1-q/2}. \tag{8}$$

Proof. The claim follows from Theorems 3 and 4 of [4], specialized to the unperturbed model, after a deterministic normalization change. That paper uses Majoranas satisfying $\{\psi_i^{\mathrm{Zlo}}, \psi_j^{\mathrm{Zlo}}\} = \delta_{ij}\mathbb{I}$. Thus $\psi_i^{\mathrm{Zlo}} = 2^{-1/2}\psi_i$, $H_N^{\mathrm{Zlo}} = 2^{-q/2}H_N$, and

$$Z_N^{\mathrm{ours}}(z) = Z_N^{\mathrm{Zlo}}(2^{q/2}z).$$

Normalized versus unnormalized trace cancels in the ratio in (8). We choose R_q so that $\{|2^{q/2}z| \le R_q^{\mathrm{Zlo}}\}$ lies inside the zero-free and second-moment region in those theorems. This gives (7)–(8) for all sufficiently large N. □

Set

$$a = \frac{q}{2} - 1 > 0, \qquad Y_N(z;g) = \frac{Z_N(z;g)}{F_N(z)}.$$

Thus $\mathbb{E}Y_N(z) = 1$, and the right-hand side of (8) is $C_q N^{-a}$. Let P_r denote the orthogonal projection in complex $L^2(\gamma_N)$ onto the r-th homogeneous Gaussian chaos, where γ_N denotes standard Gaussian measure on $\mathbb{R}^{M_N}$. We write $\|X\|_2^2 = \mathbb{E}|X|^2$. We use the probabilists' Hermite polynomials

$$H_\alpha(g) = \prod_{I \in \mathcal{I}_N} \mathrm{He}_{\alpha_I}(g_I), \qquad \mathbb{E}[H_\alpha H_\eta] = \alpha!\, \mathbf{1}_{\{\alpha=\eta\}},$$

where $\alpha! = \prod_I \alpha_I!$. For every sufficiently regular random variable X,

$$\|P_r X\|_2^2 = \sum_{|\alpha|=r} \frac{|\mathbb{E}\, \partial_g^\alpha X|^2}{\alpha!}. \tag{9}$$

Lemma 2.2. *For each fixed N, the map $z \mapsto Z_N(z;\cdot)$ is entire with values in $L^2(\gamma_N)$. Hence $z \mapsto Y_N(z;\cdot)$ is $L^2(\gamma_N)$-valued analytic on every zero-free domain of F_N.*

Proof. Since $\|\psi_I\| = 1$,

$$\|H_N(g)\| \le \sigma_N \sum_{I \in \mathcal{I}_N} |g_I|.$$

For $|z| \le R$ and every $k \ge 0$, the spectral theorem gives

$$|\partial_z^k Z_N(z;g)| \le \|H_N(g)\|^k\, e^{R\|H_N(g)\|}.$$

The square of the right-hand side is integrable because the Gaussian vector is finite dimensional. Dominated convergence therefore permits differentiation in L^2 on every compact disk. Division by the nonvanishing deterministic analytic function F_N proves the second assertion. □

2.2 Annealed free energy and coefficient asymptotics

To identify the deterministic coefficient $A_N(\beta)$ in (16), we use the annealed free-energy limit from [5]. The required convergence of derivatives follows from convexity and is not an additional input.

Lemma 2.3 (Convex convergence and derivatives)**.** *Let $I \subset \mathbb{R}$ be open. Suppose that $h_N : I \to \mathbb{R}$ are finite, differentiable, convex functions and $h_N(x) \to h(x)$ pointwise, where $h \in C^1(I)$. Then $h_N \to h$ and $h_N' \to h'$ locally uniformly on I.*

Proof. Fix a compact interval $[a,b] \subset I$ and choose $u < a < b < v$ in I. Monotonicity of secant slopes bounds every secant slope of h_N on $[a,b]$ between quantities formed from $h_N(u), h_N(a), h_N(b), h_N(v)$. These endpoint values converge, so the functions h_N are equi-Lipschitz on $[a,b]$. Pointwise convergence on a finite mesh, followed by mesh refinement, gives uniform convergence there.

For $x \in I$ and sufficiently small $s > 0$, convexity gives

$$\frac{h_N(x) - h_N(x-s)}{s} \le h_N'(x) \le \frac{h_N(x+s) - h_N(x)}{s}.$$

First let $N \to \infty$ and then $s \downarrow 0$ to obtain $h_N'(x) \to h'(x)$. Finally, h_N' are nondecreasing. On a fine partition of $[a,b]$, monotonicity traps $h_N'(x)$ between its values at adjacent mesh points, while uniform continuity traps $h'(x)$ between the corresponding values of h'. Pointwise convergence at the finitely many mesh points then gives uniform convergence of the derivatives. □

Proposition 2.4 (High-temperature annealed free energy)**.** *There are $\beta_{\mathrm{FE}} = \beta_{\mathrm{FE}}(q) > 0$ and an even real-analytic function ℓ_q on $(-\beta_{\mathrm{FE}}, \beta_{\mathrm{FE}})$ such that, locally uniformly on this interval,*

$$\frac{1}{N} \log F_N(\beta) \longrightarrow \ell_q(\beta), \tag{10}$$

$$\frac{1}{N} \partial_\beta \log F_N(\beta) \longrightarrow \ell_q'(\beta). \tag{11}$$

Moreover,

$$\ell_q(\beta) = \frac{\beta^2}{2q} + O(\beta^4), \qquad \ell_q'(\beta) = \frac{\beta}{q} + O(\beta^3). \tag{12}$$

Proof. The pointwise convergence in (10) is Theorem 5 of [5], together with the Gaussian extension stated there. That extension invokes the Gaussian–Rademacher universality result [6].

We first record the normalization conversion. The Majoranas of [5] satisfy

$$\psi_i^{\mathrm{GPSZ}} = 2^{-1/2} \psi_i, \qquad H_N^{\mathrm{GPSZ}} = 2^{-q/2} H_N.$$

Set $u = 2^{q/2}\beta$. Gaussian symmetry gives

$$\mathbb{E} \operatorname{tr} e^{-\beta H_N} = \mathbb{E} \operatorname{tr} e^{+\beta H_N},$$

and $u H_N^{\mathrm{GPSZ}} = \beta H_N$. Thus the sign and normalization conventions of [5] give

$$\frac{1}{N} \log F_N(\beta) = \frac{u}{N} f_{N,q}^{\mathrm{GPSZ}}(u), \tag{13}$$

where $f_{N,q}^{\mathrm{GPSZ}}(u) = u^{-1} \log \mathbb{E} Z_N^{\mathrm{GPSZ}}(u)$.

We next extract the regularity and the first coefficient from the explicit formula in that theorem. In its notation,

$$\Phi_u(z) = \sum_{v \ge 1} \mu_v(u) z^v, \qquad u f_q^{\mathrm{GPSZ}}(u) = \Phi_u(t_*(u)) - 1 - \log t_*(u),$$

where $t_* \partial_z \Phi_u(t_*) = 1$. A contributing hypertree has $v = 1 + (q-1)s$ vertices and its coefficient contains

$$\rho(u)^{v-1}, \qquad \rho(u)^{q-1} = \frac{(q-1)! u^2}{2^{q+1}}.$$

More explicitly, $\mathcal{T}_v^{\mathrm{sim}}$ is empty unless $v = 1 + (q-1)s$ for some $s \ge 0$. For $s \ge 1$ we may therefore write

$$\mu_{1+(q-1)s}(u) = c_{q,s} u^{2s},$$

where

$$c_{q,s} = \frac{1}{(1+(q-1)s)!} \left(\frac{(q-1)!}{2^{q+1}} \right)^s \sum_{T \in \mathcal{T}_{1+(q-1)s}^{\mathrm{sim}}} b_T.$$

The bounds used in the proof of Proposition 3 of [5] give $|c_{q,s}| \le C_q^s$. Hence

$$\Phi_u(z) = z + \sum_{s \ge 1} c_{q,s} u^{2s} z^{1+(q-1)s}$$

converges normally for (u^2, z) in a neighborhood of $(0, 1)$. Thus $\Phi_u(z)$ is jointly analytic in (u^2, z) there. At $u = 0$, $\Phi_0(z) = z$ and $t_*(0) = 1$; the derivative with respect to t of $t\partial_z\Phi_u(t) - 1$ is 1 at $(u, t) = (0, 1)$. The analytic implicit function theorem makes t_*, and hence the limiting right-hand side, analytic in u^2.

The first nontrivial term is the unique one-edge hypertree on q vertices:

$$\Phi_u(z) = z + \frac{u^2}{q2^{q+1}} z^q + O(u^4).$$

It follows successively that

$$t_*(u) = 1 - \frac{u^2}{2^{q+1}} + O(u^4), \qquad u f_q^{\mathrm{GPSZ}}(u) = \frac{u^2}{q2^{q+1}} + O(u^4).$$

Using $u = 2^{q/2}\beta$ in (13) gives (12).

It remains only to justify the two local-uniform assertions. Let $h_N = N^{-1} \log F_N$. If

$$\overline{\nu}_N(B) = \mathbb{E}\ \mathrm{tr}\, \mathbf{1}_{\{H_N \in B\}},$$

then $F_N(\beta) = \int e^{-\beta x} \overline{\nu}_N(\mathrm{d}x)$; hence h_N is finite, differentiable, and convex. Differentiation is justified by the Gaussian exponential-integrability bound in Lemma 2.2. The limiting function just obtained is C^1. Lemma 2.3 now proves both local-uniform convergences. □

By (16) and (11),

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2} \ell_q'(\beta). \tag{14}$$

3 Gaussian-chaos decomposition

For a fixed test function f, Feng–Tian–Wei [3] prove the Gaussian SYK linear-statistics CLT by isolating

$$Q_N := \mathrm{tr}\, \widehat{H}_N^2 - \mathbb{E}\, \mathrm{tr}\, \widehat{H}_N^2 = \frac{1}{M_N} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1). \tag{15}$$

In the fixed-q regime, their coefficient is

$$A(f) = \int_{\mathbb{R}} \frac{x f'(x)}{2}\, \rho_\infty(\mathrm{d}x),$$

where ρ_∞ is the limiting spectral measure. Their proof shows that the centered statistic associated with $f - A(f)x^2$ has variance $o(M_N^{-1})$. Since $\sqrt{M_N} Q_N \Rightarrow \mathcal{N}(0, 2)$, the radial term gives their limit.

There are two obstructions to substituting (6) directly into that theorem. First, $f_{N,\beta}$ depends on N through $\tau_N \asymp \sqrt{N}$. Second, neither the fixed-function polynomial approximation nor the bounded-derivative estimate in [3] is uniform for this sequence. In particular, the theorem as stated in [3] does not imply a CLT for $Z_N(\beta)$.

We instead work with the relative partition function Y_N and replace the limiting coefficient $A(f)$ by the exact finite-N coefficient

$$A_N(\beta) := \frac{\beta}{2}\, \partial_\beta \log F_N(\beta). \tag{16}$$

The next lemma proves the exact radial decomposition

$$Y_N(\beta) - 1 = A_N(\beta) Q_N + \mathcal{R}_N(\beta), \qquad \mathcal{R}_N = \sum_{r \geq 3} P_r Y_N. \tag{17}$$

The estimate replacing the fixed-test-function variance computation of [3] is

$$\mathbb{E}|\mathcal{R}_N(\beta)|^2 = o(N^2/M_N) = o(N^{2-q}). \tag{18}$$

Sections 3.2–3.4 prove precisely (18).

3.1 Exact finite-size radial projection

Lemma 3.1 (First and second chaoses)**.** *For every real β,*

$$P_1 Y_N(\beta) = 0$$

and

$$P_2 Y_N(\beta) = \frac{\beta\, \partial_\beta \log F_N(\beta)}{2M_N} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1). \tag{19}$$

Equivalently, Clifford trace orthogonality gives

$$\operatorname{tr}(\psi_I \psi_J) = \mathbf{1}_{\{I=J\}}, \qquad \operatorname{tr} H_N^2 = \sigma_N^2 \sum_I g_I^2.$$

so the second chaos is the L^2 projection of $Y_N - 1$ onto the centered radial statistic $\operatorname{tr} H_N^2 - \mathbb{E} \operatorname{tr} H_N^2$.

Proof. For $s = (s_1, \dots, s_N) \in \{\pm 1\}^N$, define $\chi_s(I) = \prod_{x \in I} s_x$. We first prove in detail that

$$Z_N(\beta; g) = Z_N\big(\beta; (\chi_s(I) g_I)_I\big). \tag{20}$$

To make the Clifford symmetry explicit, let

$$\Gamma \quad \text{be a unitary phase multiple of} \quad \psi_1 \cdots \psi_N.$$

Since N is even, moving ψ_j through the other $N-1$ Majorana generators produces the sign $(-1)^{N-1} = -1$. Hence

$$\Gamma \psi_j \Gamma^* = -\psi_j \qquad (j \in [N]).$$

For each $x \in [N]$, put $U_x = \Gamma \psi_x$. This is unitary, and conjugation by U_x is the composition of conjugation by ψ_x and conjugation by Γ. The Clifford relations give

$$\psi_x \psi_j \psi_x = \begin{cases} \psi_x, & j = x, \\ -\psi_j, & j \neq x. \end{cases}$$

Combining the last two displays, we obtain

$$U_x \psi_j U_x^* = \begin{cases} -\psi_x, & j = x, \\ \psi_j, & j \neq x. \end{cases}$$

Thus conjugation by U_x changes the sign of precisely the generator ψ_x.

Let $S_s = \{x \in [N] : s_x = -1\}$ and let U_s be the product of the unitaries U_x over $x \in S_s$, in any fixed order. Successively applying the preceding conjugation rule yields

$$U_s \psi_j U_s^* = s_j \psi_j \qquad (j \in [N]).$$

If $I = \{i_1 < \cdots < i_q\}$, then conjugation is multiplicative, so the definition $\psi_I = \mathrm{i}^{q/2} \psi_{i_1} \cdots \psi_{i_q}$ gives

$$\begin{aligned} U_s \psi_I U_s^* &= \mathrm{i}^{q/2} (U_s \psi_{i_1} U_s^*) \cdots (U_s \psi_{i_q} U_s^*) \\ &= \left(\prod_{x \in I} s_x \right) \psi_I = \chi_s(I) \psi_I. \end{aligned}$$

Writing $g^{(s)} = (\chi_s(I) g_I)_I$, we therefore have

$$\begin{aligned} H_N(g^{(s)}) &= \sigma_N \sum_{I \in \mathcal{I}_N} \chi_s(I) g_I \psi_I \\ &= \sigma_N \sum_{I \in \mathcal{I}_N} g_I U_s \psi_I U_s^* = U_s H_N(g) U_s^*. \end{aligned}$$

The power-series definition of the exponential then implies

$$e^{-\beta H_N(g^{(s)})} = U_s e^{-\beta H_N(g)} U_s^*.$$

Finally, the normalized trace is invariant under unitary conjugation. Consequently

$$Z_N(\beta; g^{(s)}) = \operatorname{tr}\Big(U_s e^{-\beta H_N(g)} U_s^*\Big) = \operatorname{tr} e^{-\beta H_N(g)} = Z_N(\beta; g),$$

which proves (20).

More precisely,

$$H_\alpha((\chi_s(I) g_I)_I) = \left(\prod_{x=1}^{N} s_x^{\sum_{I \ni x} \alpha_I}\right) H_\alpha(g).$$

Averaging the Hermite coefficient against all s shows that it vanishes unless $\sum_{I \ni x} \alpha_I$ is even for every x. This parity rule kills the first chaos. In the second chaos it also kills every off-diagonal monomial $g_I g_J$, since the parity condition for such a monomial forces $I = J$. Permutation symmetry shows that the remaining diagonal coefficients agree. Consequently

$$P_2 Z_N(\beta) = b_N(\beta) \sum_I (g_I^2 - 1)$$

for a deterministic scalar b_N.

It remains to compute the coefficient. For each fixed I, the probabilists' second Hermite polynomial is $g_I^2 - 1$, with

$$\mathbb{E}(g_I^2 - 1)^2 = 2.$$

Since P_2 is an orthogonal projection and the variables g_I are independent, taking the inner product of the preceding formula for $P_2 Z_N$ with $g_I^2 - 1$ gives

$$2 b_N(\beta) = \mathbb{E}\big[Z_N(\beta)(g_I^2 - 1)\big].$$

Applying Gaussian integration by parts twice, conditionally on all coordinates other than g_I, yields

$$\mathbb{E}\big[Z_N(\beta)(g_I^2 - 1)\big] = \mathbb{E}\, \partial_{g_I}^2 Z_N(\beta).$$

Summing over the M_N choices of I therefore gives the first equality below. A further conditional Gaussian integration by parts gives the second:

$$\begin{aligned} 2 M_N b_N(\beta) &= \sum_I \mathbb{E}\, \partial_{g_I}^2 Z_N(\beta) = \sum_I \mathbb{E}\big[g_I \partial_{g_I} Z_N(\beta)\big] \\ &= \beta\, \partial_\beta \mathbb{E} Z_N(\beta) = \beta F_N'(\beta). \end{aligned}$$

The last equality follows pointwise from the exact scaling identity

$$Z_N(\beta; tg) = \operatorname{tr} e^{-\beta H_N(tg)} = \operatorname{tr} e^{-t\beta H_N(g)} = Z_N(t\beta; g),$$

where we used $H_N(tg) = t H_N(g)$. Differentiating at $t = 1$ gives

$$\sum_I g_I \partial_{g_I} Z_N(\beta; g) = \beta\, \partial_\beta Z_N(\beta; g).$$

Finally,

$$\partial_\beta Z_N(\beta; g) = -\operatorname{tr}\Big(H_N(g) e^{-\beta H_N(g)}\Big), \qquad |\partial_\beta Z_N(\beta; g)| \le \|H_N(g)\|\, e^{|\beta| \|H_N(g)\|}.$$

Together with

$$\|H_N(g)\| \le \sigma_N \sum_I |g_I|,$$

the right-hand side has every exponential moment under the finite dimensional Gaussian law. It gives a locally uniform dominating function in β. The same type of bound justifies the preceding Gaussian integrations by parts. Thus differentiation passes through the expectation and $\partial_\beta \mathbb{E} Z_N(\beta) = F_N'(\beta)$. Dividing by the deterministic quantity $F_N(\beta)$ proves (19). □

3.2 Analytic control of the high-order remainder

The next lemma upgrades the polynomially weak estimate (8) to the precision required for the CLT.

Lemma 3.2 (Analytic suppression of high chaoses)**.** *Let $w \in \mathbb{C}$ satisfy $|w| < R$, and let $G(z)$ be analytic on a neighborhood of $|z| \le R$ with values in a Hilbert orthogonal sum $\mathcal{H} = \bigoplus_{r\ge 1} \mathcal{H}_r$. Write $G_r(z)$ for its projection onto $\mathcal{H}_r$. Suppose that G_r has a zero of order at least r at the origin. Then, for every integer $m \ge 0$,*

$$\sum_{r>m} \|G_r(w)\|_{\mathcal{H}}^2 \le \frac{(|w|/R)^{2(m+1)}}{1-(|w|/R)^2} \frac{1}{2\pi} \int_0^{2\pi} \left\|G(Re^{i\theta})\right\|_{\mathcal{H}}^2 \, d\theta. \tag{21}$$

Proof. Write

$$G_r(z) = \sum_{k\ge r} z^k v_{r,k}, \qquad v_{r,k} \in \mathcal{H}_r.$$

Parseval's identity in the angular variable gives

$$A_r(R) := \frac{1}{2\pi} \int_0^{2\pi} \left\|G_r(Re^{i\theta})\right\|_{\mathcal{H}}^2 \, d\theta = \sum_{k\ge r} R^{2k} \|v_{r,k}\|_{\mathcal{H}}^2 .$$

With $t = |w|/R < 1$, Cauchy–Schwarz yields

$$\|G_r(w)\|_{\mathcal{H}} \le \left(\sum_{k\ge r} t^{2k}\right)^{1/2} \left(\sum_{k\ge r} R^{2k} \|v_{r,k}\|_{\mathcal{H}}^2\right)^{1/2}.$$

Consequently,

$$\|G_r(w)\|_{\mathcal{H}}^2 \le \left(\sum_{k\ge r} t^{2k}\right) A_r(R) = \frac{t^{2r}}{1-t^2} A_r(R).$$

Sum over $r > m$, use $t^{2r} \le t^{2(m+1)}$, and then use orthogonality of the spaces $\mathcal{H}_r$ under the angular integral. □

Lemma 3.3 (Vanishing order)**.** *For every $r \ge 1$, the $L^2(\gamma_N)$-valued analytic function $z \mapsto P_r Y_N(z)$ has a zero of order at least r at $z = 0$.*

Proof. The Taylor coefficient of order k of $Z_N(z; g)$ is

$$\frac{(-1)^k}{k!} \operatorname{tr} H_N(g)^k,$$

which is a polynomial of total degree k in the Gaussian vector g. It has no Gaussian-chaos component of order larger than k. Therefore $P_r Z_N(z)$ is divisible by z^r. Since F_N is deterministic, $F_N(0) = 1$, and F_N has no zero on the disk under consideration, division by $F_N(z)$ preserves this vanishing order. □

Apply Lemma 3.2 to $G(z) = Y_N(z) - 1$ and $G_r(z) = P_r Y_N(z)$. Combining Proposition 2.1 with Lemmas 3.2 and 3.3, for every $0 < |\beta| < R < R_q$ and every $m \ge 0$ we obtain

$$\sum_{r>m} \|P_r Y_N(\beta)\|_2^2 \le \frac{C_q}{1-(|\beta|/R)^2} N^{-a} \left(\frac{|\beta|}{R}\right)^{2(m+1)}. \tag{22}$$

3.3 Uniform control of chaoses up to logarithmic order

Lemma 3.4 (Real-temperature derivative bound)**.** *For real β, every multi-index α of total order r satisfies*

$$\left|\partial_g^\alpha Z_N(\beta;g)\right| \le (|\beta|\sigma_N)^r Z_N(\beta;g). \tag{23}$$

Proof. Label the r differentiations by $I_1,\dots,I_r$. The iterated Duhamel formula is a sum over $r!$ permutations of integrals over the simplex $\Delta_r = \{t_j \ge 0 : \sum_{j=0}^r t_j = 1\}$, whose volume is $1/r!$. Every integrand, apart from the factor $(|\beta|\sigma_N)^r$, has the form

$$\operatorname{tr}\left(e^{-t_0\beta H_N}\psi_{I_{\pi(1)}}e^{-t_1\beta H_N}\cdots\psi_{I_{\pi(r)}}e^{-t_r\beta H_N}\right).$$

Here and below the Schatten norms are defined using the normalized trace:

$$\|A\|_p = (\operatorname{tr}|A|^p)^{1/p}, \qquad \|A\|_\infty = \|A\|.$$

For $t_j > 0$,

$$\left\|e^{-t_j\beta H_N}\right\|_{1/t_j} = \left(\operatorname{tr} e^{-\beta H_N}\right)^{t_j},$$

with the usual $p=\infty$ interpretation when $t_j = 0$. Noncommutative Hölder, with Schatten exponents $1/t_j$, and $\|\psi_I\|_\infty = 1$, therefore bounds the absolute value of the integrand by

$$\prod_{j=0}^r \left(\operatorname{tr} e^{-\beta H_N}\right)^{t_j} = Z_N(\beta;g).$$

The permutation count cancels the simplex volume, proving (23). The same labeled formula covers repeated coordinates. □

Call a multi-index α *even* if

$$\sum_{I\ni x}\alpha_I \equiv 0 \pmod 2 \qquad \text{for every } x\in[N].$$

The sign-gauge symmetry used in Lemma 3.1 shows that only even multi-indices have nonzero Hermite coefficients.

Lemma 3.5 (Logarithmic-order finite-chaos estimate)**.** *For every fixed real β there is $D = D(q,\beta) < \infty$ such that, for all $r \ge 1$ and all N,*

$$\|P_r Y_N(\beta)\|_2^2 \le \exp\big(Dr\log(r+1)\big)N^{-ar}. \tag{24}$$

Consequently, for every fixed $A < \infty$,

$$\sum_{r=3}^{\lceil A\log N\rceil} \|P_r Y_N(\beta)\|_2^2 = o(N^{-2a}). \tag{25}$$

Proof. The Hermite coefficient formula, Lemma 3.4, and $F_N(\beta) > 0$ give

$$\begin{aligned}\|P_r Y_N(\beta)\|_2^2 &= \frac{1}{F_N(\beta)^2}\sum_{|\alpha|=r}\frac{1}{\alpha!}\left|\mathbb{E}\partial_g^\alpha Z_N(\beta)\right|^2 \\ &\le (|\beta|\sigma_N)^{2r}\sum_{\substack{|\alpha|=r\\ \alpha\text{ even}}}\frac{1}{\alpha!}.\end{aligned} \tag{26}$$

An even ordered r-tuple of q-sets has total incidence qr, and every vertex in its union has incidence at least two. Its union therefore has at most $qr/2$ vertices. Since a multi-index α corresponds to

$r!/\alpha!$ ordered tuples, we may first choose a v-set containing the union and then choose the ordered tuple. This can count a tuple more than once, which is harmless for the upper bound. Thus

$$\sum_{\substack{|\alpha|=r\\ \alpha \text{ even}}} \frac{1}{\alpha!} \le \frac{1}{r!} \sum_{v \le qr/2} \binom{N}{v} \binom{v}{q}^r \le \left(\frac{qr}{2}+1\right) \left(\frac{qr}{2}\right)^{qr} N^{qr/2}.$$

Using $\sigma_N^{2r} = ((q-1)!)^r N^{-(q-1)r}$ in (26) proves (24) after increasing D.

Let $m_N = \lceil A \log N \rceil$. Uniformly for $3 \le r \le m_N$, $D \log(r+1) = o(\log N)$. Hence, for all sufficiently large N,

$$\exp\big(Dr \log(r+1)\big) N^{-ar} \le N^{-3ar/4}.$$

Summing from $r = 3$ to m_N gives $O(N^{-9a/4}) = o(N^{-2a})$. □

3.4 Completion of the higher-chaos estimate

Proposition 3.6 (Higher-chaos closure)**.** *For every fixed real β satisfying $0 < |\beta| < R_q$,*

$$\sum_{r \ge 3} \|P_r Y_N(\beta)\|_2^2 = o(N^{2-q}) = o(N^{-2a}). \tag{27}$$

Proof. Choose R with $|\beta| < R < R_q$ and put $t = |\beta|/R < 1$. Choose a constant A such that

$$2A|\log t| > a,$$

and set $m_N = \lceil A \log N \rceil$. Lemma 3.5 controls $3 \le r \le m_N$. For the remaining chaoses, (22) gives

$$\sum_{r > m_N} \|P_r Y_N(\beta)\|_2^2 \le \frac{C_q}{1-t^2} N^{-a} t^{2(m_N+1)} = o(N^{-2a}).$$

Adding the two estimates proves the claim. □

Proof of Theorem 1.1. Let

$$S_N = \frac{1}{\sqrt{2M_N}} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1).$$

The classical i.i.d. central limit theorem gives $S_N \Rightarrow \xi \sim \mathcal{N}(0,1)$. By Lemma 3.1,

$$Y_N(\beta) - 1 = c_N(\beta) S_N + \mathcal{R}_N(\beta), \qquad c_N(\beta) = \frac{\beta\, \partial_\beta \log F_N(\beta)}{\sqrt{2M_N}}, \tag{28}$$

where $\mathcal{R}_N = \sum_{r \ge 3} P_r Y_N$ is orthogonal to S_N. Since $M_N \sim N^q/q!$, (11) gives

$$N^{q/2-1} c_N(\beta) \longrightarrow \sqrt{\frac{q!}{2}}\, \beta \ell_q'(\beta).$$

Proposition 3.6 gives $N^{q/2-1}\mathcal{R}_N \to 0$ in L^2. Slutsky's theorem proves (1). Orthogonality in (28) also gives

$$\mathrm{Var}(Y_N) = c_N(\beta)^2 + \|\mathcal{R}_N(\beta)\|_2^2,$$

which proves (2). Finally, $\mathrm{Var}(Y_N) = \mathrm{Var}(Z_N)/(\mathbb{E}Z_N)^2$, and the assumed nonvanishing of the limiting coefficient allows one more application of Slutsky, proving (3). □

Proof of Corollary 1.2. Put

$$a = \frac{q}{2} - 1, \qquad X_N = \frac{Z_N(\beta) - \mathbb{E}Z_N(\beta)}{\mathbb{E}Z_N(\beta)}, \qquad W_N = \log \frac{Z_N(\beta)}{\mathbb{E}Z_N(\beta)}.$$

Thus $W_N = \log(1 + X_N)$, and Theorem 1.1 gives

$$N^a X_N \Longrightarrow \sqrt{\frac{q!}{2}}\, \beta \ell_q'(\beta)\, \xi, \qquad \delta_N^2 := \mathbb{E}X_N^2 \asymp N^{-2a}.$$

We claim that

$$\mathbb{E}W_N = o(N^{-a}). \tag{29}$$

To prove this, we first establish an N-independent Lipschitz bound for $g \mapsto W_N(g)$. Differentiation gives

$$\partial_{g_I} W_N = -\beta \sigma_N \langle \psi_I \rangle_{\beta,g}.$$

For each $(q-1)$-set S, the operators $(\psi_{S\cup\{i\}})_{i\notin S}$ are pairwise anticommuting Hermitian involutions. If $A_1, \dots, A_m$ are any such operators and $a_j = \langle A_j \rangle$ in a state, set $B = \sum_j a_j A_j$ and $s = \sum_j a_j^2$. Pairwise anticommutation gives

$$B^2 = s\mathbb{I}, \qquad s = \langle B \rangle \le \|B\| = \sqrt{s},$$

and hence $s \le 1$. Applying this observation for each S and noting that every $I \in \mathcal{I}_N$ is counted exactly q times yields

$$\sum_{I\in\mathcal{I}_N} |\langle \psi_I \rangle_{\beta,g}|^2 \le \frac{1}{q}\binom{N}{q-1}.$$

Consequently,

$$\sum_{I\in\mathcal{I}_N} |\partial_{g_I} W_N|^2 \le \frac{\beta^2\sigma_N^2}{q}\binom{N}{q-1} \le \frac{\beta^2}{q}.$$

Writing $\mu_N = \mathbb{E}W_N$, the Gaussian concentration inequality therefore implies, for every $\lambda \in \mathbb{R}$,

$$\mathbb{E}e^{\lambda(W_N - \mu_N)} \le \exp\left(\frac{\lambda^2\beta^2}{2q}\right). \tag{30}$$

Jensen's inequality and the identity $\mathbb{E}e^{W_N} = 1$ give

$$-\frac{\beta^2}{2q} \le \mathbb{E}W_N \le 0.$$

Indeed, the upper bound follows from Jensen, while (30) with $\lambda = 1$ gives the lower bound.

Let $K_\beta = \beta^2/(2q)$. Since $e^{W_N} - 1 = X_N$, Chebyshev's inequality gives, for $t \ge 1$,

$$\mathbb{P}(W_N < -t) \le \frac{\delta_N^2}{(1 - e^{-t})^2} \le C\delta_N^2.$$

On the other hand, $\mu_N \ge -K_\beta$, so (30) and the Chernoff bound imply

$$\mathbb{P}(W_N < -t) \le \exp\left(-\frac{q(t - K_\beta)^2}{2\beta^2}\right), \qquad t > K_\beta.$$

Set

$$T_N = \max\left\{1,\, K_\beta + \frac{2|\beta|}{\sqrt{q}}\sqrt{\log(e/\delta_N)}\right\}.$$

Indeed, the layer-cake formula gives

$$\mathbb{E}\left[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\right] = \mathbb{P}(W_N < -1) + \int_1^\infty \mathbb{P}(W_N < -t)\,\mathrm{d}t.$$

The interval $[1, T_N]$ contributes at most $C\delta_N^2 T_N$, while integration of the Gaussian bound over $[T_N, \infty)$ contributes at most $C\delta_N^2$. Consequently,

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le C\delta_N^2\sqrt{\log(e/\delta_N)}. \tag{31}$$

Now set $h(w) = e^w - 1 - w$. There is an absolute constant C such that

$$h(w) \le C(e^w-1)^2 \quad (w \ge -1), \qquad h(w) \le |w| \quad (w < -1).$$

Since $\mathbb{E}e^{W_N} = 1$, (31) yields

$$0 \le -\mu_N = \mathbb{E}h(W_N) \le C\delta_N^2\sqrt{\log(e/\delta_N)} = O\Big(N^{2-q}\sqrt{\log N}\Big) = o(N^{-a}).$$

This proves (29).

It remains to compare the random parts. The sequence $N^a X_N$ is tight and $X_N \to 0$ in probability. On the event $\{|X_N| \le 1/2\}$, the Taylor bound $|\log(1+x) - x| \le Cx^2$ gives

$$N^a|W_N - X_N| \le C\big(N^a|X_N|\big)|X_N|.$$

The right-hand side converges to zero in probability, while $\mathbb{P}(|X_N| > 1/2) \to 0$. Therefore

$$N^a\big(W_N - X_N\big) \longrightarrow 0 \qquad \text{in probability.}$$

Together with (29), this shows that

$$N^a\big((W_N - \mathbb{E}W_N) - X_N\big) \longrightarrow 0 \qquad \text{in probability.}$$

Finally,

$$\log Z_N - \mathbb{E}\log Z_N = W_N - \mathbb{E}W_N,$$

so the claimed convergence follows from (1) and Slutsky's theorem. The assertion for $\mathcal{G}_N$ follows by multiplication by $-\beta^{-1}$. □

Remark 3.7. Since $\ell_q'(\beta) = \beta/q + O(\beta^3)$ at the origin, the nondegeneracy condition in Theorem 1.1 holds for every sufficiently small real $\beta \neq 0$.

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